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(a)(i)	$P = \rho gh$ $= 1030 \times 9.81 \times 400$ $= 4.04 \text{ MPa (3 s.f.)}$
(a)(ii)	$\sigma = \frac{d_0 P}{2t}$ $t \geq \frac{d_0 P}{2\sigma} \geq \frac{6.30 \times 4.04 \times 10^6}{2 \times 550 \times 10^6}$ $t \geq 0.0231 \text{ m}$ $= 2.4 \text{ cm (round up to 2 s.f.) (shown)}$
(a)(iii)	The thickness of the hull is much smaller than the diameter/radius of the submarine.
(a)(iv)	$E = \frac{\sigma}{\frac{d_x}{d_0}}$ $d_x = \frac{\sigma d_0}{E} = \frac{550 \times 10^6 \times 6.30}{205 \times 10^9}$ $d_x = 16.9 \text{ mm}$
	New diameter = $6.30 - (16.9 \times 10^{-3}) = 6.28 \text{ m (3 s.f.)}$

(a)(v)	A heavier submarine with a thicker hull has reduced manoeuvrability and slower speeds due to increased drag / requires higher power with a bigger engine. OR A thicker hull translates to a heavier submarine which requires a higher ballast capacity for buoyancy. This reduces room for crew and equipment. OR Building extremely thick hulls requires specialized materials and techniques, making them more expensive and time-consuming to manufacture.
(b)(i)	Sound waves are longitudinal waves OR The displacement of the water particles/ molecules of the medium is parallel to (or along) the direction of wave propagation
	The transfer of energy is via changing pressure (varying rarefactions and compressions).
(b)(ii)	$c = 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3$ $+ (1.34 - 0.010T)(S - 35) + 0.016D$ $c = 1449.2 + 4.6(28) - 0.055(28)^2 + 0.00029(28)^3$ $+ (1.34 - 0.010(28))\left(\frac{33}{1000} - 35\right) + 0.016(400)$
	$c = 1500 \text{ m s}^{-1}$ (2 s.f.)
(b)(iii)	Water is a less compressible medium / higher density medium as compared to air. OR Water molecules are at a closer proximity that enables faster transfer of vibrational energy between the molecules.
(b)(iv)	$I \text{ at M} = \frac{P}{S_{\perp}} = \frac{P}{4\pi d^2}$
	$P \text{ at M} = \frac{P}{4\pi d^2} S$ Reflected $P \text{ at M} = \frac{P}{4\pi d^2} S$
	Reflected $I \text{ from M} (I_M) = \frac{PS}{4\pi d^2} \frac{1}{\frac{4\pi d^2}{2}} = \frac{PS}{8\pi^2 d^4}$
	$d = \left[\frac{PS}{(I_M)8\pi^2} \right]^{\frac{1}{4}}$ (shown)
(c)(i)	$P = Fv = \frac{1}{2} C_d \rho v^2 A v$ $P = \frac{1}{2} (0.1)(1030) \left(\frac{37.0 \times 1000}{60 \times 60} \right)^3 \left(\frac{\pi (6.30)^2}{4} \right)$
	$P = 1.74 \times 10^6 \text{ W}$ (3 s.f.)

(c)(ii)	The submarine includes irregular shaped features such as control tower, fins and antennas that increases drag coefficient/area and hence increasing the power required. OR The flow of sea water over the submarine may turbulent, hence increasing the drag coefficient and the increasing power required. OR The hull of the submarine is not smooth ie covered by barnacles, hence increasing the drag coefficient and increasing the power required. OR The density of seawater is increased at other regions and hence more power is required. OR The efficiency of the engine is not 100%. Hence more power is required.
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